**Student's Name……..Signature……..**

**525/1**
**CHEMISTRY**
(Paper 1)
**April**
**2026**
2hrs; 30mins

**MATEETE COMPREHESIVE SEED SECONDARY SCHOOL**

**END OF TERM ONE ASSESSMENT 2026**

Uganda Advanced Certificate of Education

**CHEMISTRY**
(Theory)
**SENIOR FIVE**

**2hours 30minutes**

**INSTRUCTIONS TO CANDIDATES:**

*This paper has two sections A and B*

*Section A has two* ***compulsory items*** *while B has two parts Part 1 and Part II. Each of part I and part II has one item and it is* ***compulsory***

*Responses to section A* ***must*** *be written in spaces provided and section B on the answer sheet provided. Use blue or black ink and a pencil for drawings*

*Mathematical tables, silent non-programmable scientific calculators, and graphs may be used*

***Where necessary use,***

*Molar gas volume at s.t.p =22.4dm$^3$, Molar gas constant R=8.31JK$^{-1}$mol$^{-1}$, Standard pressure=101325Nm$^{-2}$,standard temperature=273K and* ***Periodic table at the end of paper***

| For Scorer's Use Only | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **ITEM** | **1** | | | | | **2** | | | | | | **3** | | | | **4** | | | | **TOTAL** |
| **CODE** | | | | | | | | | | | | | | | | | | | | |
| **SCORE** | | | | | | | | | | | | | | | | | | | | |

## ***SECTION A (compulsory)***

**ITEM 1**

James a senior technician in masala sec school went to lab technician to buy some sulphuric acid. On reaching the school, he discovered the following information on the bottle, molecular weight; w/w 98%, specific gravity (density) $1.84gcm^{-1.}$ This information could not lead him directly to the actual volume required to prepare the standard solution.

**TASK**

(a) As a chemistry student, using the knowledge of mole concept, advise her on how he could prepare 0.25M solution of sulphuric acid for 50 students given that each student is to receive $200cm^3$.

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(b) The students are to use the above standardized solution to determine the percentage purity of anhydrous sodium carbonate using the information below. 17.6 grams of impure sodium carbonate was dissolved in $500cm^3$ of distilled water to make a solution. $25cm^3$ of this solution required $28.5cm^3$ of 0.25M sulphuric acid for complete reaction. Determine;

i. Moles of sulphuric acid that reacted.

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ii. Moles of sodium carbonate that reacted.

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iii. Moles of sodium carbonate in 500cm$^3$ of solution.

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iv) Mass of sodium carbonate in 500cm$^3$ of solution hence the percentage purity of sodium carbonate.

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**ITEM 2**

A diagnostic laboratory receives a blood sample from a patient suspected of heavy-metal exposure. Mass spectrometry of the sample shows distinct peaks for several lead isotopes. The international level of concern for lead in blood is $4.8 \times 10^{-7}$ mol $dm^{-}3$. The patient's blood test reports a concentration of $1.80 \times 10^{-4}$ g dm-3 of lead. To determine whether this result exceeds the international guideline, the laboratory analyst must calculate the relative atomic mass (RAM) of lead from the mass spectrum and then convert the patient's mass-based value into moles per litre (mol $L^{-1}$) for comparison.

| Isotope | m/z | Relative Abundance (%) |
|---|---|---|
| $^{204}Pb$ | 204 | 1.4% |
| $^{206}Pb$ | 206 | 24.1% |
| $^{207}Pb$ | 207 | 22.1% |
| $^{208}Pb$ | 208 | 52.4% |

**TASK**

a) Explain the existence of isotopes and how they give rise to different spectral lines.

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b) Determine whether the patient's symptoms are caused by the lead poisoning.

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## ***SECTION B (compulsory)***

**ITEM 3**

An industrial chemist wanted to analyze the trend in the atomic radius of sodium, magnesium, aluminum, phosphorous and chlorine with electronic configuration 11, 12,13,15 and 17 respectively and the size of ions formed by atoms using an X-ray diffractor. During the analysis, she obtained with the following results;

| Element | Na | Mg | Al | P | Cl |
|---|---|---|---|---|---|
| Atomic radius (nm) | 0.156 | 0.136 | 0.125 | 0.110 | 0.099 |
| Ionic radius (nm) | 0.023 | 0.020 | 0.010 | 0.222 | 0.235 |

The industrial chemist after obtaining the result was confused by some atoms forming bigger ions than the corresponding while others forming smaller ions than the corresponding atoms. And she also noted that atomic radius was decreasing from sodium to chlorine became inquisitive about the factors that could have caused that.

**TASK**

As a chemistry student,

a) Plot graphs of atomic radius and ionic radius against element on the same axis
b) write a help to interpret the data obtained and account for the differences and observations made. (include the electronic configurations where necessary)

**ITEM 4**

A chemical research laboratory in Uganda is studying a gaseous hydrocarbon X obtained from a petrochemical source. Elemental analysis shows that X contains 11.11 % hydrogen by mass, and its vapour density is 27, giving a relative molecular mass of 54. When 1 mol of X reacts with 1 mol of hydrogen over a nickel catalyst at 150 °C, the product still contains a carbon–carbon double bond. The laboratory wishes to identify X, investigate its reactions, and highlight safe handling and environmental precautions when working with gaseous hydrocarbons.

**TASK**

As a Chemistry student, help the laboratory by :

(a) Determine the molecular formula of X and state its class of hydrocarbons.

b i) Write all the possible isomers of X

ii ) Explain the types of isomerism shown by compound X and provide examples in each case from the isomers you drew in b (i)

c i) It was analysed that X has a functional group on carbon atom 2, identify X and describe reaction of the alkene product of X with hydrogen bromide

ii) Outline to the laboratory techinicians the mechanism for the reaction above

(c) State one danger of handling gaseous hydrocarbons and how it can be mitigated



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## Periodic Table

| 1 | 2 | | | | | | | | | | | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.0 H 1 | | | | | | | | | | | | | | | | 1.0 H 1 | 4.0 He 2 |
| 6.9 Li 3 | 9.0 Be 4 | | | | | | | | | | | 10.8 B 5 | 12.0 C 6 | 14.0 N 7 | 16.0 O 8 | 19.0 F 9 | 20.2 Ne 10 |
| 23.0 Na 11 | 24.3 Mg 12 | | | | | | | | | | | 27.0 Al 13 | 28.1 Si 14 | 31.0 P 15 | 32.1 S 16 | 35.4 Cl 17 | 40.0 Ar 18 |
| 39.1 K 19 | 40.1 Ca 20 | 45.0 Sc 21 | 47.9 Ti 22 | 50.9 V 23 | 52.0 Cr 24 | 54.9 Mn 25 | 55.8 Fe 26 | 58.9 Co 27 | 58.7 Ni 28 | 63.5 Cu 29 | 65.7 Zn 30 | 69.7 Ga 31 | 72.6 Ge 32 | 74.9 As 33 | 79.0 Se 34 | 79.9 Br 35 | 83.8 Kr 36 |
| 85.5 Rb 37 | 87.6 Sr 38 | 88.9 Y 39 | 91.2 Zr 40 | 92.9 Nb 41 | 95.9 Mo 42 | 98.9 Tc 43 | 101 Ru 44 | 103 Rh 45 | 106 Pd 46 | 108 Ag 47 | 112 Cd 48 | 115 In 49 | 119 Sn 50 | 122 Sb 51 | 128 Te 52 | 127 I 53 | 131 Xe 54 |
| 133 Cs 55 | 137 Ba 56 | 139 La 57 | 178 Hf 72 | 181 Ta 73 | 184 W 74 | 186 Re 75 | 190 Os 76 | 192 Ir 77 | 195 Pt 78 | 197 Au 79 | 201 Hg 80 | 204 TI 81 | 207 Pb 82 | 209 Bi 83 | 209 Po 84 | 210 At 85 | 222 Rn 86 |
| 223 Fr 87 | 226 Ra 88 | 227 Ac 89 | | | | | | | | | | | | | | | |
| | | | 139 La 57 | 140 Ce 58 | 141 Pr 59 | 144 Nd 60 | 147 Pm 61 | 150 Sm 62 | 152 Eu 63 | 157 Gd 64 | 159 Tb 65 | 162 Dy 66 | 165 Ho 67 | 167 Er 68 | 169 Tm 69 | 173 Yb 70 | 175 Lu 71 |
| | | | 227 Ac 89 | 232 Th 90 | 231 Pa 91 | 238 U 92 | 237 Np 93 | 244 Pu 94 | 243 Am 95 | 247 Cm 96 | 247 Bk 97 | 251 Cf 98 | 254 Es 99 | 257 Fm 100 | 256 Md 101 | 254 No 102 | 260 Lw 103 |



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